PMx Weekly Digest: November 01, 2026

Content Source: Full Text

Paper 1: Longitudinal Modeling of Cardiometabolic Dysfunction in Patients With Type 2 Dia

Bibliographic Information

- **Title**: Longitudinal Modeling of Cardiometabolic Dysfunction in Patients With Type 2 Diabetes Using Item Response Theory.
- **Authors**: Chengli Yu, Hanna Kunina, Gustaf J. Wellhagen, Maria C. Kjellsson
- **Journal**: CPT: Pharmacometrics & Systems Pharmacology
- **DOI**: 10.1002/psp4.70349

Paper Type: METHODOLOGY

Quick Take

This study applies multidimensional item response theory (MIRT) to routinely collected cardiometabolic biomarkers in 33,396 newly diagnosed type 2 diabetes patients, identifying two latent dimensions (blood pressure and metabolic) that capture complementary aspects of cardiometabolic dysfunction. The longitudinal MIRT model with an item-specific slope for HbA1c describes distinct trajectories and shows modest predictive value for all-cause mortality beyond age and sex.

Executive Summary

The authors develop a longitudinal graded-response multidimensional IRT (MIRT) model to characterize cardiometabolic dysfunction in patients with type 2 diabetes (T2D). Using data from the Swedish National Diabetes Register (33,396 patients, 915,218 measurements), they categorize seven biomarkers (sBP, dBP, LDL, HDL, TG, HbA1c, BMI) into ordered categories and demonstrate that a two-dimensional model (blood pressure-related and metabolic latent variables) fits substantially better than a unidimensional model. The model reveals that blood pressure items dominate the BP dimension, while metabolic items (HDL, HbA1c, TG, BMI) inform the metabolic dimension, with LDL showing low discrimination. Longitudinal analysis shows modest declines in both latent variables, with HbA1c requiring an item-specific slope to capture its distinct trajectory. The latent variables show modest predictive value for all-cause mortality, supporting the utility of this multidimensional approach.

Scientific Context & Motivation

The paper addresses the limitation of traditional approaches that treat cardiometabolic risk factors as independent predictors, failing to capture the shared variance and multidimensional nature of cardiometabolic dysfunction. It proposes a multidimensional item response theory (MIRT) framework to integrate multiple biomarkers into latent measures of disease severity, enabling longitudinal assessment of disease progression across multiple domains.

⚡ Methodological Snapshot

The study employs a graded-response multidimensional item response theory (MIRT) model to analyze ordered categorical biomarker data. The model estimates latent variables (ψ_1 and ψ_2) that represent underlying cardiometabolic dysfunction dimensions. The analysis proceeds in steps: dimensionality assessment using AIC/BIC, baseline model estimation with covariates, longitudinal extension with time-varying latent variables and item-specific slopes, and survival analysis using the latent variables as predictors. The model is implemented in NONMEM, with dimensionality assessment in R using the mirt package.

⊠ Statistical Framework

The graded response model (GRM) assumes that the probability of a patient scoring at or above a category threshold on an item follows a logistic function of the latent trait(s).[1] The model assumes local independence, meaning that item responses are independent given the latent variables.[2] The multidimensional extension allows items to load on multiple latent dimensions, with discrimination parameters indicating the strength of association. The model uses maximum likelihood estimation, with covariates affecting the latent variables via linear regression. The longitudinal model assumes linear change in latent variables over time, with potential item-specific slopes to capture differential trajectories.

Estimator Behavior

The model parameters are estimated using maximum likelihood, which is is asymptotically unbiased and efficient under correct model specification. The large sample size (33,396 patients) ensures stable estimates with low relative standard errors (RSEs) for most parameters.[3] The discrimination parameters for LDL and TG have high RSEs (12.7-20.7%), indicating low precision,, likely due to low information from these items. The item-specific slope for HbA1c ($0.45\ year^{-1}$) is well estimated (RSE 4.6%). The model convergence is not explicitly discussed, but the use of NONMEM suggests standard convergence criteria were applied.

Validation Design

The model was validated using visual predictive checks (VPCs) for individual items and an overall percentage score VPC. The VPCs compare observed and simulated percentiles of score distributions over time, with 95% confidence intervals from simulations. The dimensionality assessment used AIC and BIC to compare models with 1-7 dimensions. The survival analysis used likelihood

[1] **UNSUPPORTED** — original: "The graded response model (GRM) assumes that the probability of a patient scoring at or above a category threshold on an item follows a logistic function of the latent trait(s)." ⟶ correction: "[flagged / unverified — no source-supported correction available]"

[2] **UNSUPPORTED** — original: "The model assumes local independence, meaning that item responses are independent given the latent variables." ⟶ correction: "[flagged / unverified — no source-supported correction available]"

[3] **CONTRADICTED** — original: "The large sample size (33,396 patients) ensures stable estimates with low relative standard errors (RSEs) for most parameters." ⟶ correction: "LDL | 0.18 (12.7) | 0.13 (13.8)"

ratio tests, C-index, and Brier score to evaluate predictive performance. No external validation dataset was used.[4]

Applicability Boundaries

The method works well when biomarkers can be categorized into ordered categories with clinical relevance, and when the latent structure is multidimensional. It is applicable to large observational datasets with repeated measurements. The method may not work well when biomarkers are continuous and categorization discards too much information, or when the latent structure is not well-defined. The model assumes linear change over time, which may not hold for all biomarkers. The method requires sufficient data to estimate multiple discrimination and difficulty parameters per item, which may be challenging with sparse data.

Comparison to Alternatives

Compared to unidimensional IRT, the multidimensional model provides a better fit and captures complementary dimensions of cardiometabolic dysfunction. Compared to confirmatory factor analysis (CFA), IRT directly models the probability of response categories and provides item-level information, allowing for more flexible modeling of ordinal data. Compared to traditional risk factor models (e.g., UKPDS-OM), this approach integrates multiple biomarkers into latent variables, accounting for shared variance. The main advantage is the ability to characterize multidimensional disease progression, while the disadvantage is increased complexity and the need for categorization.

Implementation Guidance

The model is implemented in NONMEM 7.5.1, with data management in R. The dimensionality assessment uses the mirt package in R. For practical implementation, users need to categorize continuous biomarkers into ordered categories based on clinical guidelines. The model requires specification of the number of dimensions, item discrimination and difficulty parameters, and covariate effects. The longitudinal model requires defining latent variable trajectories and potential item-specific slopes. Computational cost is moderate, with the large dataset requiring efficient estimation methods. The use of VPCs is recommended for model validation.

☒ Key Findings

A two-dimensional MIRT model identified a blood pressure-related latent variable (ψ_1) and a metabolic latent variable (ψ_2). ψ_1 was primarily informed by sBP and dBP, while ψ_2 was informed by HDL, HbA1c, TG, and BMI; LDL showed low discrimination on both dimensions. Baseline age at T2D diagnosis had opposing effects on the latent variables (0.02 for ψ_1, -0.07 for ψ_2), and sex was associated with ψ_2 (women scored 0.43 units lower than men). Longitudinally, both latent variables showed post-baseline offsets (-0 for ψ_1, -0.22 for ψ_2) and annual slopes

[4] **UNSUPPORTED** — original: "No external validation dataset was used." → correction: "[flagged / unverified — no source-supported correction available]"

of approximately $-0.03\ year^{-1}$. HbA1c exhibited a distinct trajectory with an initial post-baseline improvement followed by deterioration, requiring an item-specific slope of $0.45\ year^{-1}$. The metabolic latent variable contributed modest additional information for all-cause mortality beyond age and sex.

Strengths & Limitations

Strengths

- Large real-world dataset (33,396 patients) with comprehensive longitudinal follow-up (915,218 measurements)
- Demonstrates clear advantage of multidimensional IRT over unidimensional IRT ($\Delta AIC \approx -6000$) in AIC/BIC)
-
- Identifies two complementary latent dimensions (BP-related and metabolic) with distinct biomarker contributions
- Handles ordered categorical data via graded response model, aligning with clinical risk categories
-
- Provides a flexible framework for integrating multiple biomarkers into latent disease severity measures

Limitations (Acknowledged by Authors)

- Observational data may introduce bias due to non-random missingness and measurement errors
- Categorization of continuous biomarkers discards information and reduces sensitivity to within-category changes
- Latent variables remain abstract constructs not directly linked to clinical outcomes
- Large sample size may lead to statistically significant but clinically irrelevant differences

Limitations (Expert Review)

- The model does not account for the fact that the latent variables are not directly observable, which may limit clinical interpretation
- The study does not address the potential for informative observation bias, where patients with more severe disease may have more frequent measurements
- The model does not account for the fact that the latent variables are not directly observable, which may limits clinical interpretation
- The model does not account for the potential for informative observation bias, where patients with more severe disease may have more frequent measurements

Generalizability

The findings are based on a large Swedish registry cohort and may not generalize to other populations or healthcare settings. The model's applicability to other datasets requires validation, and the clinical utility of the latent variables needs further investigation.

Key Equations

Graded Response Model (GRM)

$$P(X_{ij} \geq k \mid \theta_i) = \frac{1}{1 + \exp\left[-a_j(\theta_i - b_{jk})\right]}$$

Probability that patient i scores at least category k on item j, given latent trait θ_i, with discrimination a_j and difficulty b_{jk}.

Figures & Tables

- **Figure 1**: Distribution of scores for all items at baseline (A) and over time (B).
  - *Significance*: Shows the prevalence of risk factor burden and temporal trends, highlighting HbA1c's distinct pattern.
- **Figure 2**: Item Characteristic Curves (ICCs) for all items from the baseline graded-response MIRT model.
  - *Significance*: Illustrates how each biomarker relates to the two latent dimensions, supporting the interpretation of the dimensions.
- **Figure 3**: Item information plots showing Fisher Item Information Curves and percentage contribution per dimension.
  - *Significance*: Quantifies the relative importance of each biomarker for each latent dimension, guiding interpretation.
- **Figure 4**: VPC of HbA1c with common shared slope (A) and separate slope (B).
  - *Significance*: Demonstrates the need for an item-specific slope to capture HbA1c's unique trajectory.
- **Figure 5**: Percentage Score VPC of the final longitudinal MIRT model.
  - *Significance*: Validates the overall model performance in describing the total cardiometabolic burden over time.

Code & Reproducibility Assessment

The analysis used NONMEM 7.5.1, PsN 5.3.1, R 4.3.3, and the mirt and piraid packages. No public code repository is mentioned, limiting full reproducibility.

Supplementary Materials

Supplementary materials include Figure S1 (AIC/BIC for dimensionality), Figure S5 (item information for unidimensional model), Table S4 (survival analysis metrics), and Table S6 (item-specific difficulty parameters).

Future Directions

Further investigations should focus on the clinical interpretation of each latent dimension through associations with specific clinical outcomes, such as cardiovascular events or microvascular com-

plications. Additionally, exploring continuous-response IRT approaches that retain continuous biomarker data could improve sensitivity to within-category changes.

Expert Commentary

This paper represents a thoughtful application of multidimensional IRT to a complex clinical problem. The methodological rigor is commendable, particularly the use of model comparison criteria and visual predictive checks. However, the clinical utility of the latent variables remains to be demonstrated, and the potential for informative observation bias is a concern. The framework could be extended to incorporate time-varying covariates and to validate against hard clinical endpoints.

Bottom Line

This study demonstrates the feasibility of using longitudinal IRT to integrate multiple biomarkers into latent measures of cardiometabolic dysfunction in patients with T2D. The two-dimensional model identified a blood pressure-related latent variable (ψ_1) and a metabolic latent variable (ψ_2), with ψ_1 primarily informed by sBP and dBP, and ψ_2 informed by HDL, HbA1c, TG, and BMI. LDL showed low discrimination on both dimensions. Age at T2D diagnosis was associated with both latent variables, with opposing effects of 0.02 for ψ_1 and -0.07 for ψ_2. Sex was associated with ψ_2, with women scoring 0.43 units lower than men. Longitudinally, ψ_1 and ψ_2 showed post-baseline offsets of -0.10 and -0.22, respectively, followed by separately estimated annual slopes of approximately $-0.03\ year^{-1}$. HbA1c exhibited a distinct trajectory with an initial post-baseline improvement followed by deterioration, requiring an item-specific slope of $0.45\ year^{-1}$. The ψ_2 contributed modest additional information for all-cause mortality beyond age and sex.

Fact-check corrections

Figures

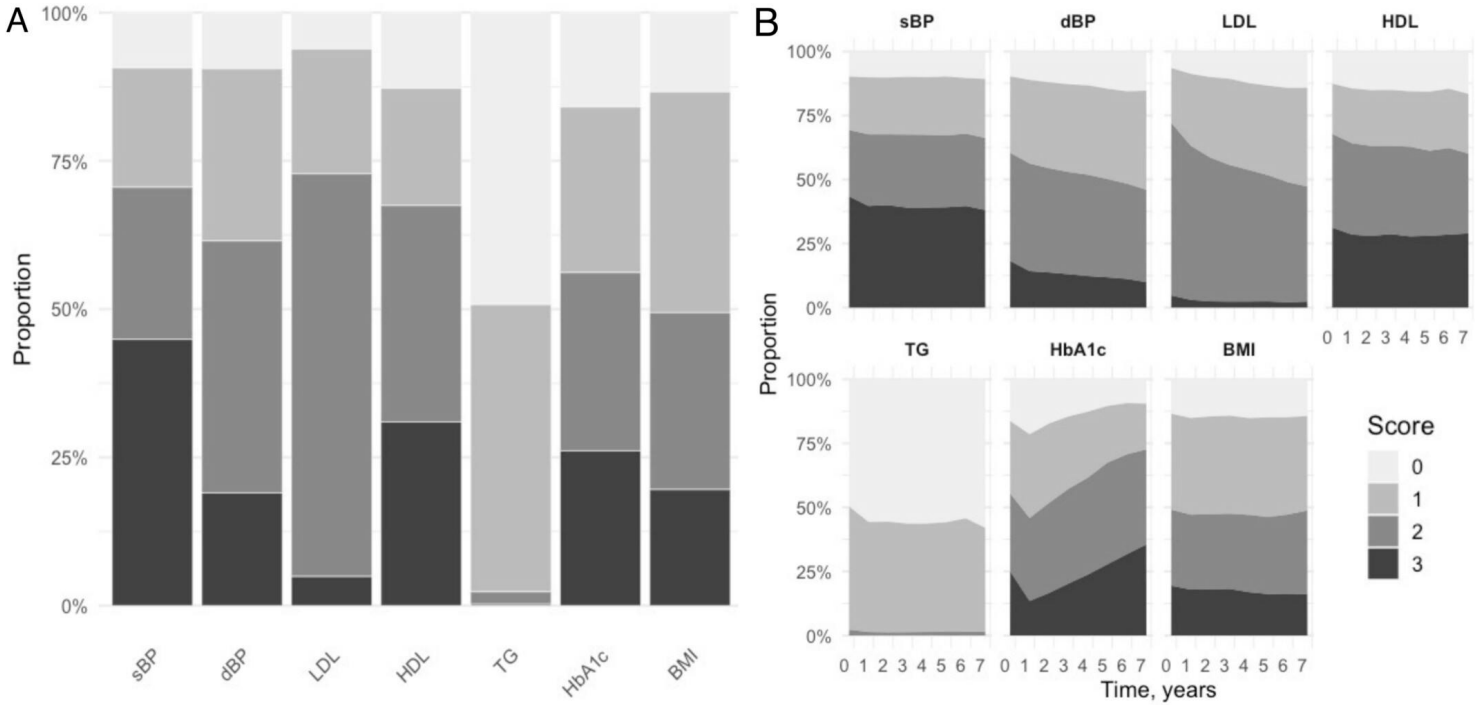


Figure 1: Distribution of scores for all items at baseline (A) and over time (B). Items: sBP—systolic blood pressure, dBP—diastolic blood pressure; LDL—low-density lipoproteins, HD

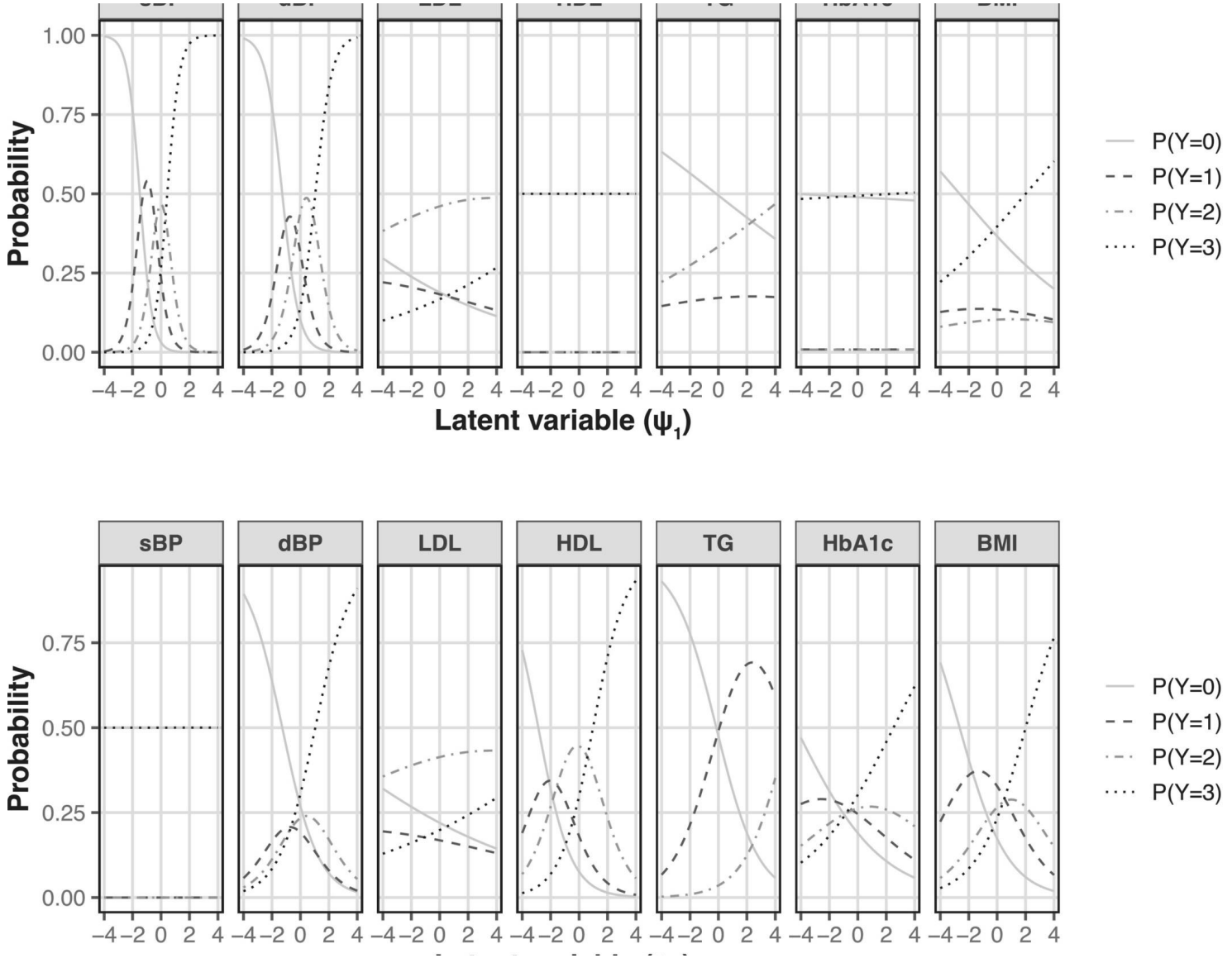


Figure 2: Item Characteristic Curves (ICCs) for all items from the baseline graded-response MIRT model. Each panel shows the model-predicted probability of falling into each of the

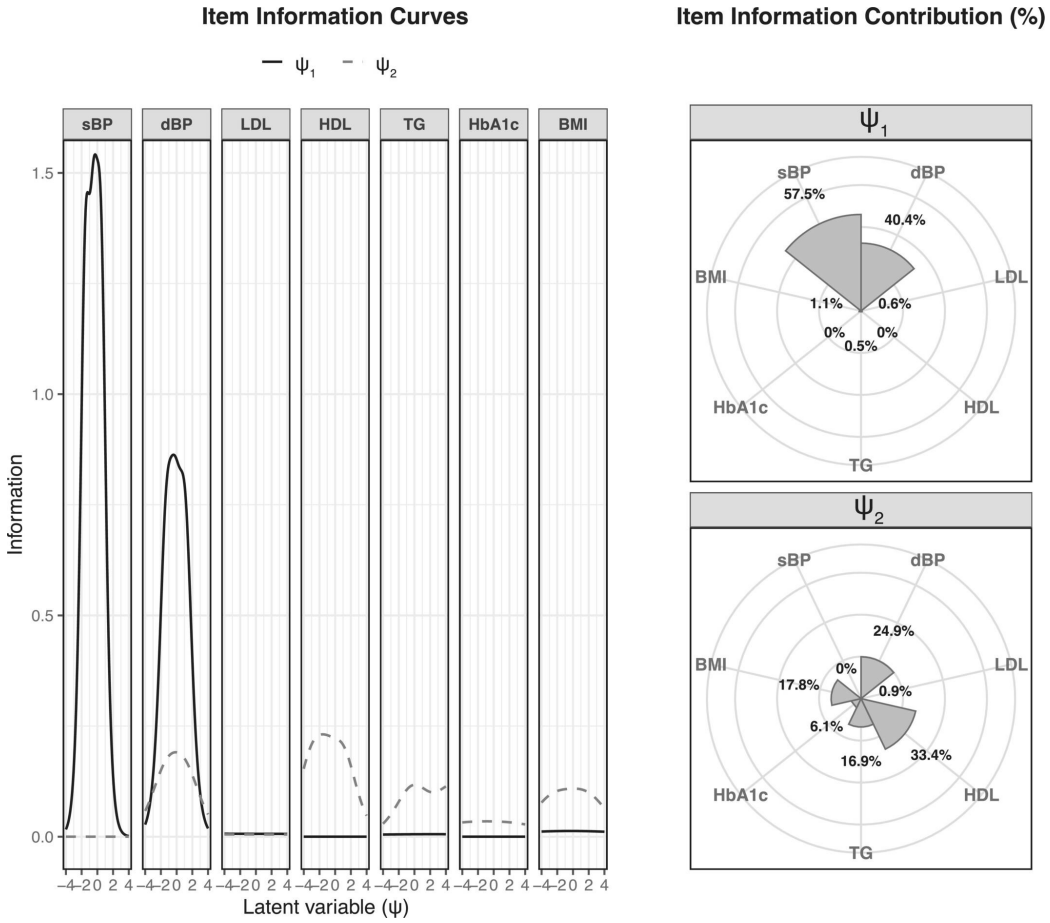


Figure 3: Item information plots for all items for the baseline graded-response MIRT model. Left panel: Fisher Item Information Curves (IICs) plot, showing how much information eac

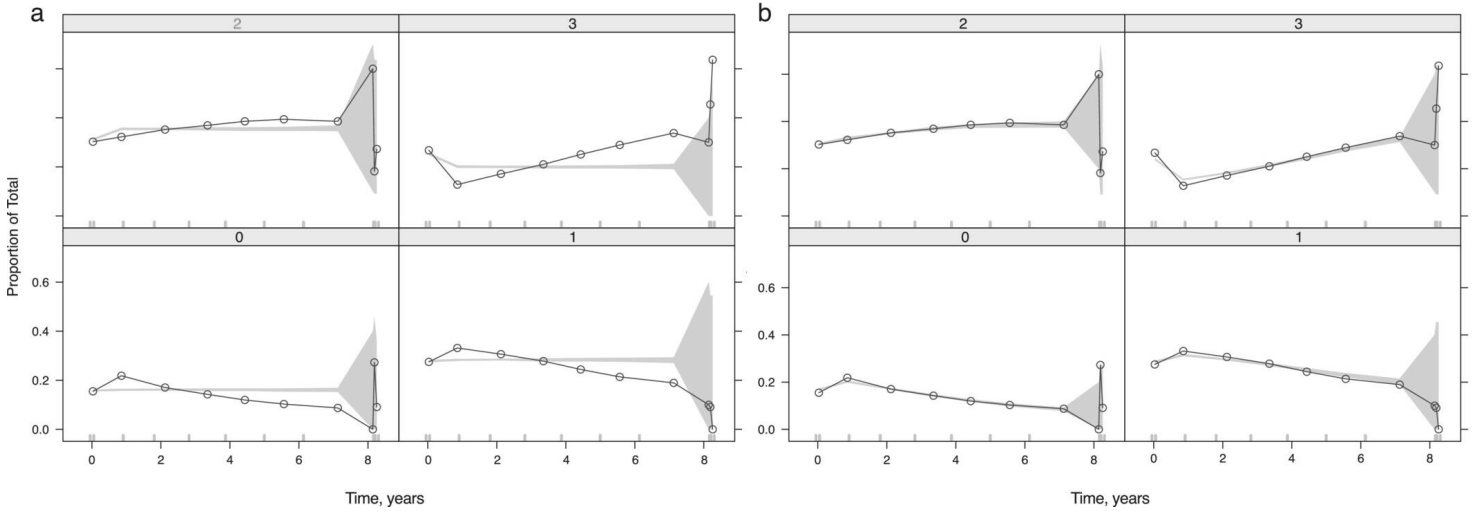


Figure 4: Visual predictive check (VPC) of HbA1c with the common shared(left panel, A) and with the separate(right panel, B). Each sub-panel shows the model-predicted probability o

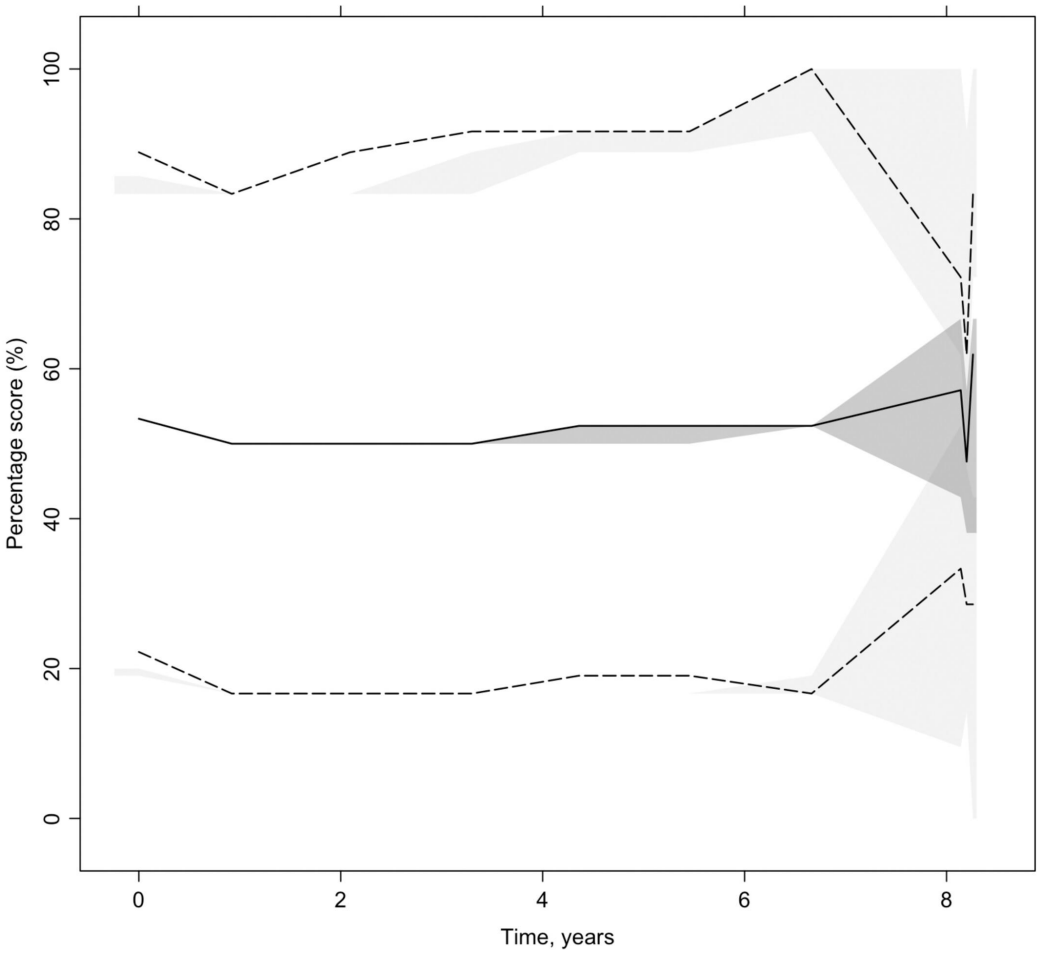


Figure 5: Percentage Score VPC of the final longitudinal MIRT model for the full study population. The lines correspond to the 97.5th (dashed), 50th (solid), and 2.5th (dashed) per